25900962 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCSDV100 - FUNDAMENTALS OF ACOUSTICS

(2024 ADMISION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer Questions

Answer any **10** Questions

Each Question carries **2** marks

1. what is the amplitude of an sound wave ? [U] / [CO1]
2. How does sound travel differently in air compared to water? [An] / [CO3]
3. How does diffraction help us hear sounds from different directions? [A] / [CO4]
4. Why are critical bands important in sound perception? [K] / [CO6]
5. What is the speed of sound in Air in relation with temperature ? [U] / [CO1]
6. what will happen if two sound waves are In' In phase' ? [An] / [CO1]
7. Give an example of a complex sound and briefly describe its characteristics [K] / [CO2]
8. Describe the relationship between density and elasticity. [An] / [CO3]
9. What are the crest and trough of a transverse wave? [U] / [CO1]
10. What is sound absorption? [U] / [CO5]

11. What happens when two sound waves meet and interfere constructively? [U] / [CO4]

12. what is frequency domin masking? [K] / [CO6]

[10x2 = 20]

Part B

Short Answer Questions

Answer any **6** Questions

Each Question carries **5** marks

13. expalin frequency range of humen hearing [U] / [CO6]

14. Compare the effects of too much and too little sound reflection in an auditorium. [An] / [CO4]

15. What is a wave Equation ? [U] / [CO1]

16. Describe how the attack and decay phases of a sound envelope can influence the perception of rhythm and dynamics in music. [An] / [CO2]

17. how we can sound proof a room ? [A] / [CO5]

18. Compare how sound propagates in air versus water. [An] / [CO4]

19. How does frequency affect the perception of pitch, and what is the importance of this in speech communication? [U] / [CO6]

20. What is pitch ? How frequency related with pitch ? [An] / [CO1]

21. what is the function of eustachian tube? [U] / [CO6]

[6x5 =30]

Part C

Essay Questions

Answer any **2** Questions

Each Question carries **10** marks

22. What are the characteristics of a sound wave . exlplain with illustaration ? [U] / [CO1]

23. What are the factors that influence acoustic impedance, and how do these factors affect the efficiency of sound transmission? [An] / [CO5]

24. Describe the relationship between amplitude, loudness, and the decibel scale, and explain how they are used to measure and perceive sound. [A] / [CO6]

[2x10=20]